

By: AKH
 Rev.--- By: ---
 Chk: ---

Date May 2026
 Date ---
 Date ---

Macaulay Test

Macaulay Beam Analysis - BetaTesting

09 - Steel - Cantilevered End (1)

Bridge, L = in
 Beam Sect = W14x61
 A = 17.90 in²
 I = 640.00 in⁴

E = 29,000,000 lb / in²
 G = 1.740E+06 lb / in² = 0.06*29,000,000

Length Units in
 Force Units lb

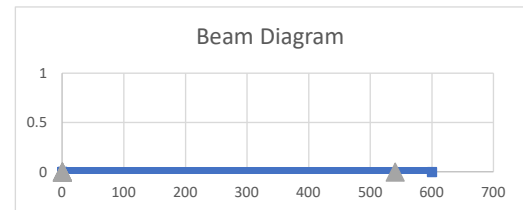
Beam Segments (First Segment Begins at 0)

	Segment No.	X End	A	I	E	G
		in	in ²	in ⁴	lb / in ²	lb / in ²
DO NOT DELETE ROWS (GRAPHED)	1	600.00	17.90	640.00	2.900E+07	3.115E+07
	2					
	3					
	4					
	5					

Supports

Perscribed Disp. (typ 0)

	Supp. No.	X	dY (vert)	dr (rot)
		in	in	rad
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.000
	2	540.0	0.0	0.000
	3			
	4			
	5			



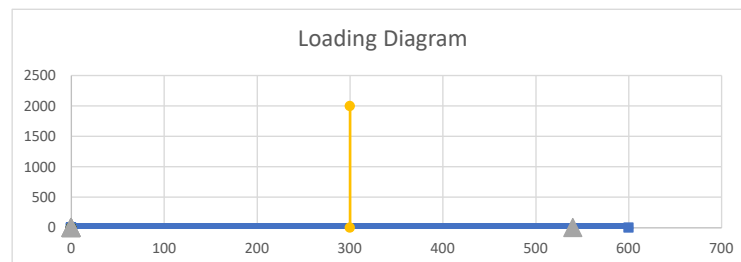
Distributed Loads (downward loads entered as NEGATIVE) - UNFACTORED

	Load No.	x1	x2	w1 (at x1)	w2 (at x2)
		in	in	lb / in	lb / in
DO NOT DELETE ROWS (GRAPHED)	1	0.0	0.0	0.0	0.0
	2	0.0	0.0	0.0	0.0
	3	0.0	0.0	0.0	0.0
	4	0.0	0.0	0.0	0.0
	5	0.0	0.0	0.0	0.0

Load
lb / ft
0.0
0.0
0.0
0.0

Point Loads - UNFACTORED

	Load No.	X	Py	My
		in	lb	lb-in
DO NOT DELETE ROWS (GRAPHED)	1	300	-2000.00	0.00
	2	0	0.00	0.00
	3			
	4			
	5			



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Macaulay Test
BEAM REACTIONS

Position (X)	Enforced Disp (dy)	Enforced Rot (theta)	Vert Reac (Ry)	Internal Reac (Mz)	Note
(L)	(L)	(rad)	(F)	(F-L)	(---)
0.00	0.0	0.0	888.9	0.0	Support reaction
540.00	0.00	0.00	1,111.1	0.0	Support reaction

Vert Reac
(kip)
0.889
1.111

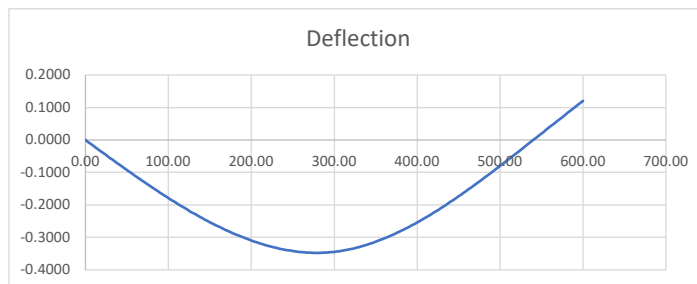
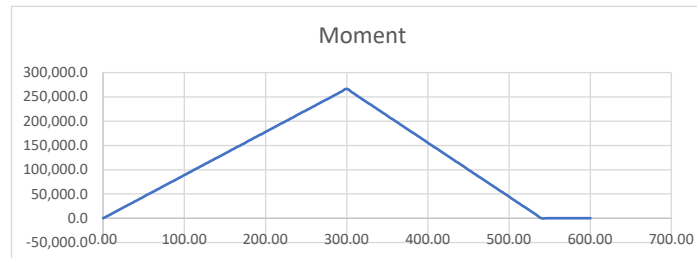
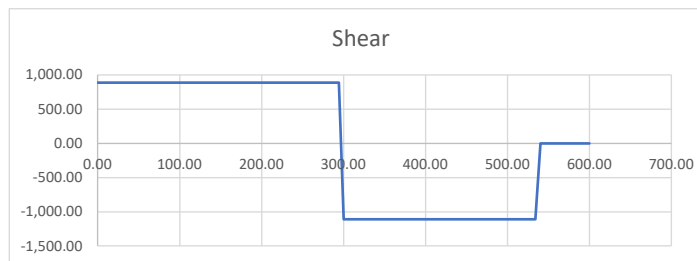
BEAM REACTIONS EQUILIBRIUM CHECK

Item	Units	Loads	Reactions	Residual	Check Sum
Sum Fy	F	-2,000.0	2,000.0	0.0	OK
Sum M @ x=0	F-L	-6.0000E+05	6.0000E+05	0.0	OK

BEAM MAX/MIN LOAD EFFECTS

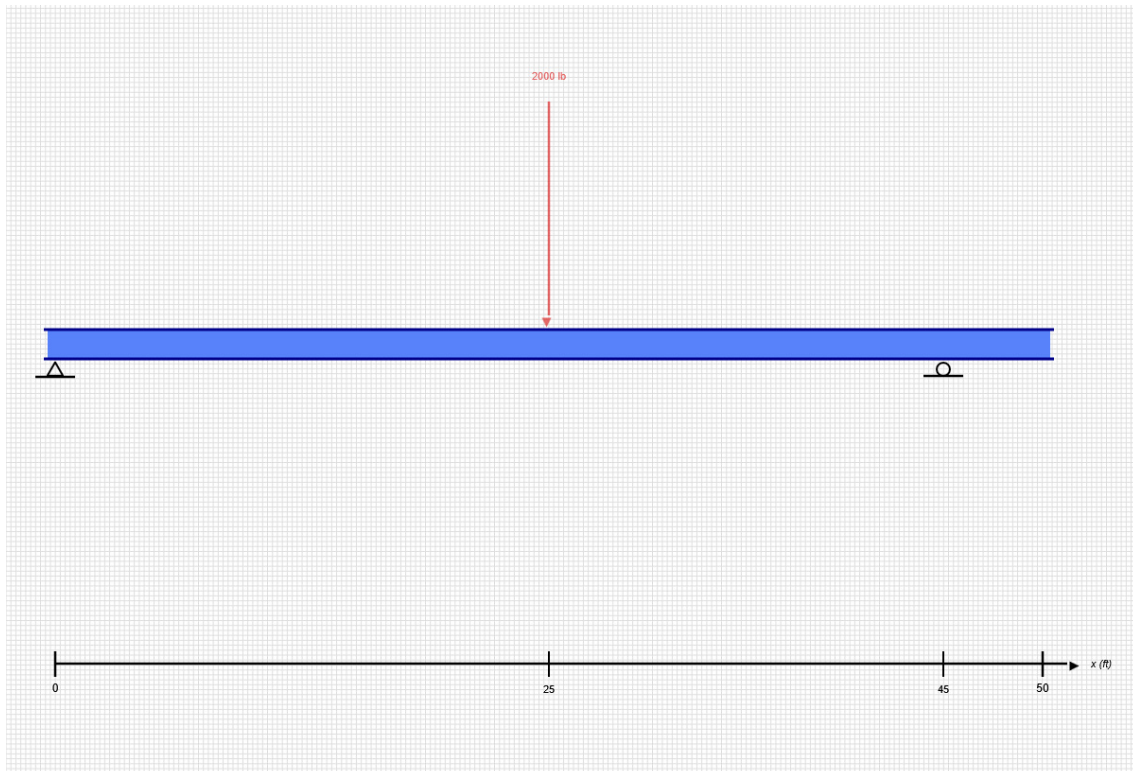
Item	Position (X)	Value
(---)	(L)	(F or F-L)
Max +V	0.00	888.9
Max -V	300.00	-1,111.1
Max +M	300.00	266,666.7
Max -M	0.00	0.0
Max Defl	282.00	-0.348

Value
(---)
0.89 kip
-1.11 kip
22.22 kip-ft
0.00 kip-ft
-0.348 in



SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2
Wed Jun 10 2026 09:17:54 GMT-0400 (Eastern Daylight Time)

Project Info

File Name: 01-Steel-Beam

Engineer: Andrew N. DeBear (debear@arninis.com)

Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results
- Bending Moment Diagram (BMD)
- Shear Force Diagram (SFD)
- Deflection Results

INPUT SUMMARY

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

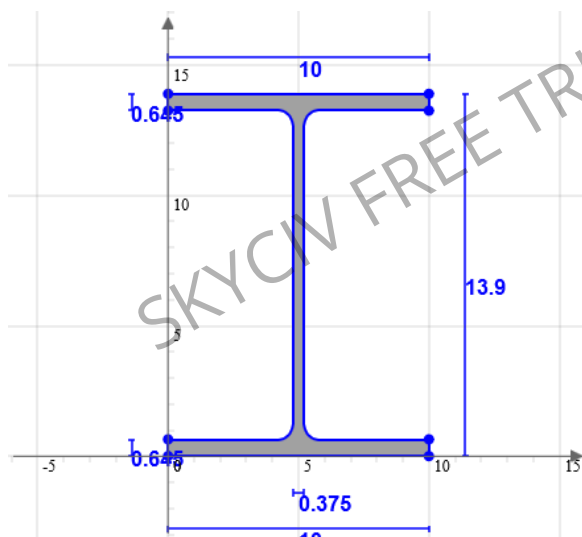
Supports

Support Type	Location
Pinned	0 ft
Roller	45 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	25 ft	-2000 lb	DL

Beam Section



Geometric Properties		
A	17.9	in ²
C _z	5	in
C _y	6.95	in

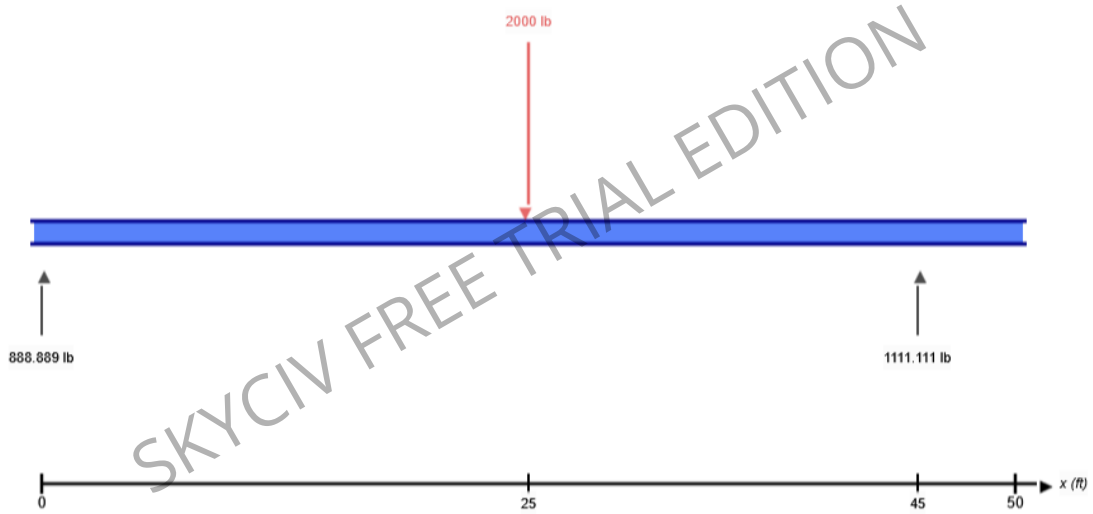
Bending Properties		
I _z	640	in ⁴
I _y	107	in ⁴

Shear Properties		
A _z	11.277	in ²
A _y	4.99	in ²

Torsion Properties		
J	2.19	in ⁴
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft ³)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.503	L/497
Custom Stress Limit	PASS	39 ksi	0.074	2.896 ksi
Material Yield	PASS	36 ksi	0.08	2.896 ksi
Material Strength	PASS	58 ksi	0.05	2.896 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	888.889	0
45	0	1111.111	0

Force Extremes

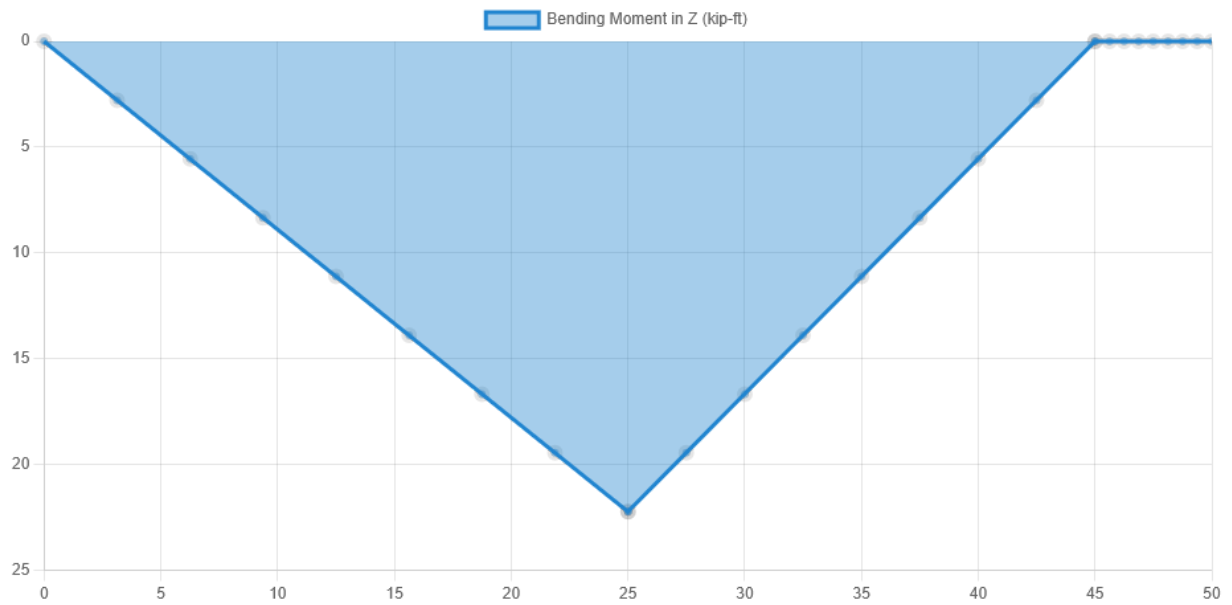
Result	Max	Min
Bending Moment	22.222 kip-ft	0 kip-ft
Shear	888.889 lb	-1111.111 lb
Displacement Y	0.121 in	-0.346 in

Stress Extremes

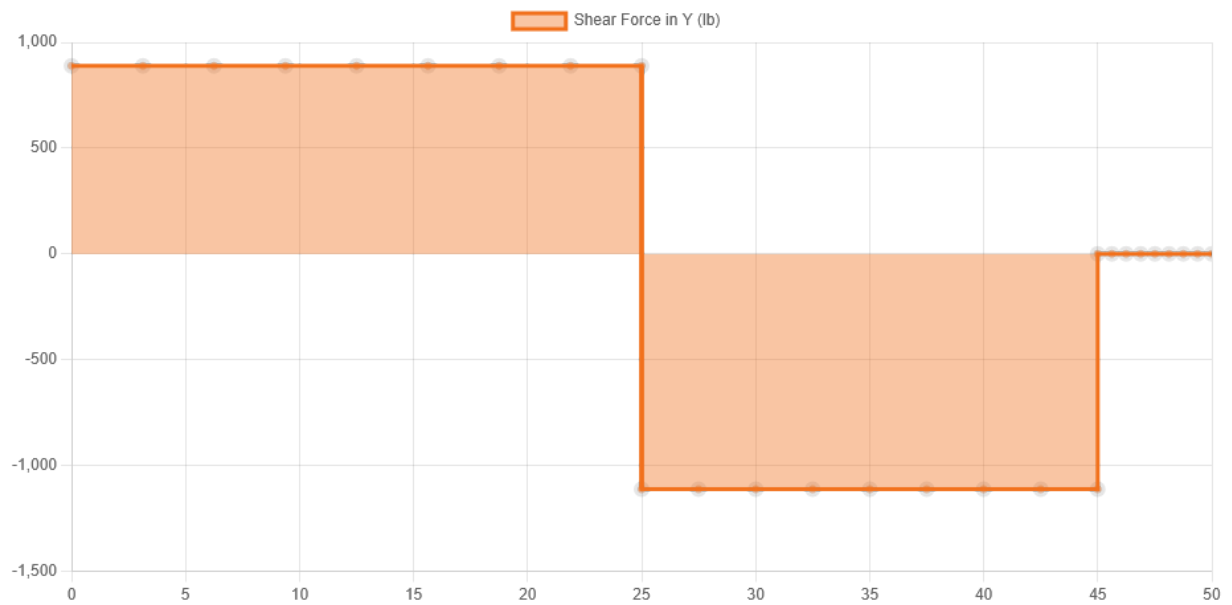
Result	Max (ksi)	Min (ksi)
Bending Stress	2.896	-2.896
Shear Stress Total	0.237	0
Max Combined Normal Stress	2.896	0
Min Combined Normal Stress	0	-2.896

DIAGRAMS

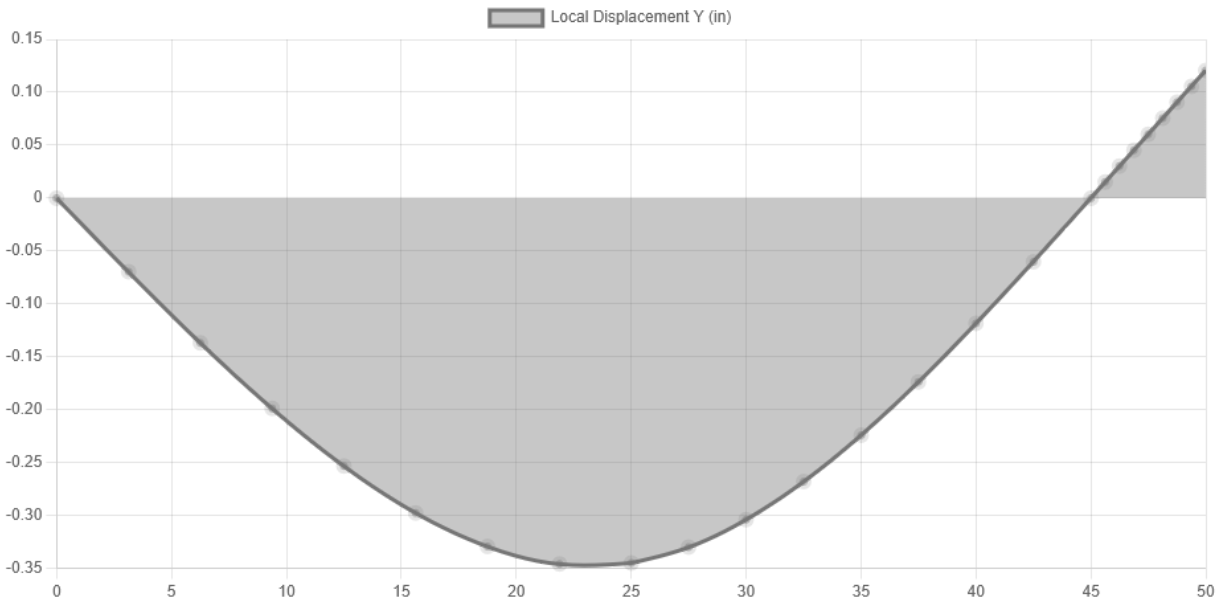
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	0	21.875	45	50
Total Deflection (in)	0 ⓘ	0.346 ⓘ	0 ⓘ	0.121 ⓘ
Deflection/Span	-	L/1561	-	L/497
ⓘ The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the model settings.				